Lab #2
Quantum Teleportation

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1. Quantum Teleportation of Unentangled Qubit

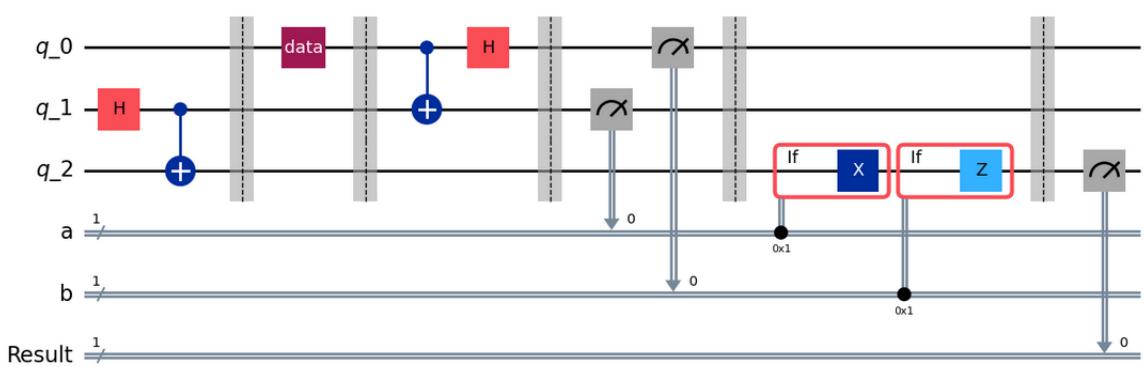
First, you will perform analysis of quantum teleportation of an unentangled qubit. The teleported qubit state is

$$|\psi\rangle = (HT)^2 HSH|0\rangle,$$

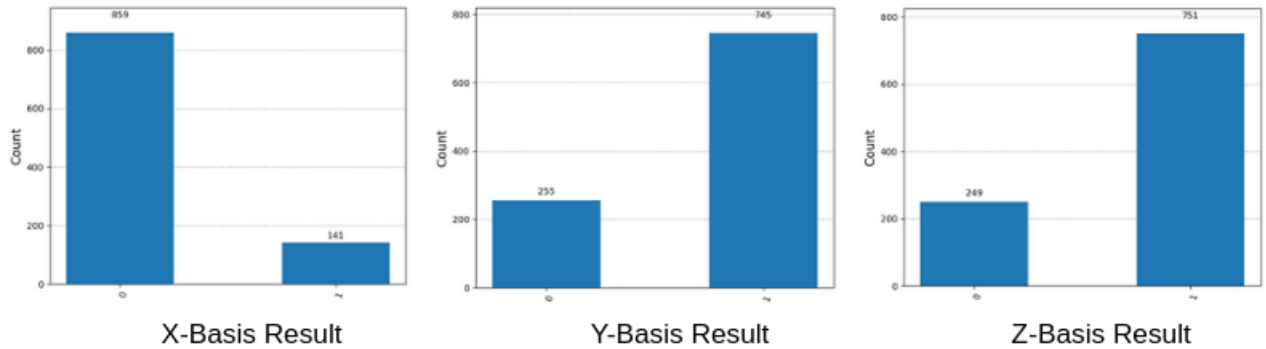
which is the same state as you used in Lab 1. You will teleport this state from qubit q_0 to qubit q_2.

First, we must define our data gate as shown below:

Next, we set up the Quantum Teleportation circuit scheme (inserting barriers to prevent the compiler from removing the teleportation) to transfer this data from q_0 to q_2:



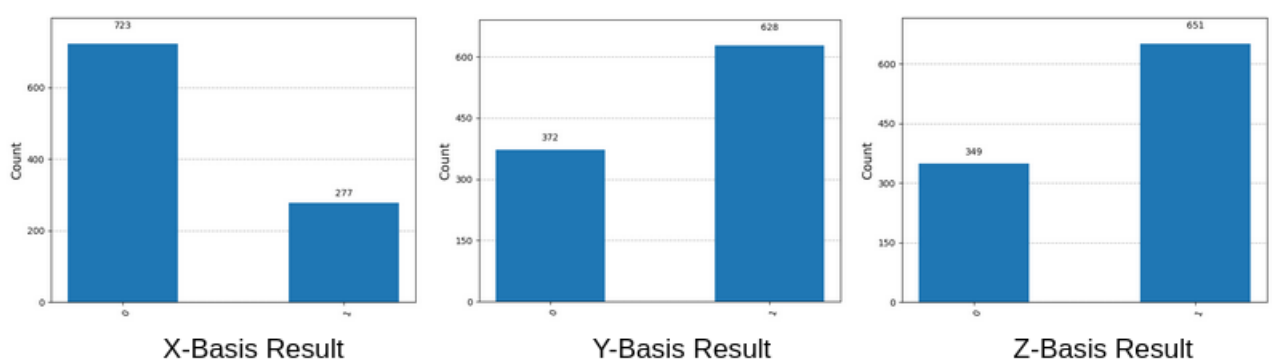
This scheme measures q_2 in the Z-basis and stores the result in a classical register. We can now simulate the results of the X-basis, Y-basis, and Z-basis measurements using the AER simulator. The plotted results are:



Thus we have theoretically predicted that q_2 will be in the 'ON' state for each of the basis measurements with the following probabilities:

$$P_X(|-\rangle) = \frac{\langle 1|\psi\rangle}{1000} = \frac{141}{1000} = 14.1\%$$
$$P_Y(|-i\rangle) = \frac{\langle 1|\psi\rangle}{1000} = \frac{745}{1000} = 74.5\%$$
$$P_Z(|1\rangle) = \frac{\langle 1|\psi\rangle}{1000} = \frac{751}{1000} = 75.1\%$$

which aligns with the results for the same state from lab #1 with a greatest difference of 0.5%. Now that we have theoretically shown that we have succesfully teleported the data, we can run the same process on a Quantum Processor. The resulting measurements were:



with the probabilities:

$$P_X(|-\rangle) = \frac{\langle 1|\psi\rangle}{1000} = \frac{277}{1000} = 27.7\%$$
$$P_Y(|-i\rangle) = \frac{\langle 1|\psi\rangle}{1000} = \frac{628}{1000} = 62.8\%$$
$$P_Z(|1\rangle) = \frac{\langle 1|\psi\rangle}{1000} = \frac{651}{1000} = 65.1\%$$

With these probabilities, we can now conduct quantum state tomography on q_2 to calculate the length of the corresponding Bloch vector n. The n components are:

$$n_x = P_X(|+\rangle) - P_X(|-\rangle) = (72.3 - 27.7)\% = 44.6\% = 0.446$$
$$n_y = P_Y(|i\rangle) - P_Y(|-i\rangle) = (37.2 - 62.8)\% = -25.6\% = -0.256$$
$$n_z = P_Z(|0\rangle) - P_Z(|1\rangle) = (34.9 - 65.1)\% = -30.2\% = -0.302$$

And the length of n is:

$$|n| = \sqrt{0.446^2 + 0.256^2 + 0.302^2} = 0.611$$

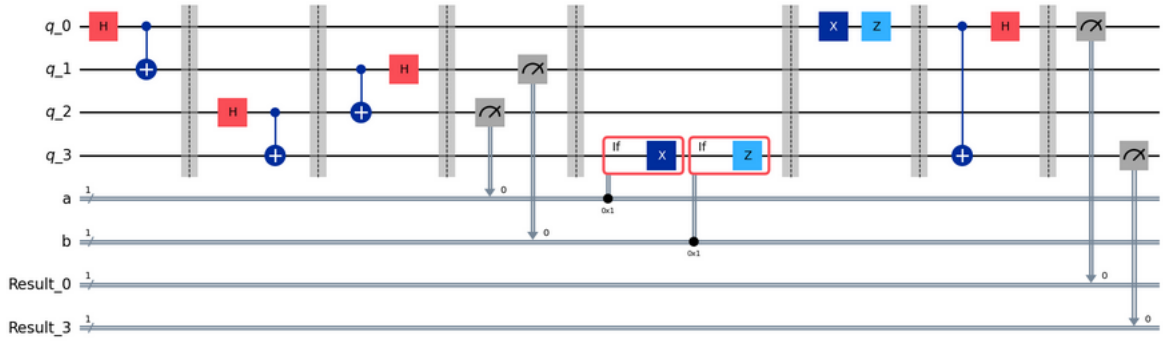
Considering that the result of the teleported state should be $|n| = 1$ based on theoretical predictions and results from lab #1, we can view this result as the fidelity of the quantum state, or how closely our teleported state aligns with the original unteleported state. In this case, our fidelity value is 61.1%, which aligns with the results of teleportation in 3-qubit IBM Quantum superconducting systems (about 60-70%)[1] and is considered quite satisfactory for an uncorrected process. This illustrates that quantum teleportation without the use of any error correction/mitigation techniques can lead to significant data loss as the information is being passed through a real quantum processing system due to factors such as noise, circuit complexity, and measurement errors. Quantum Error Correction (QEC) has shown fidelity results as high as 98.95% [2] in 3-qubit quantum teleportation schemes and emphasizes the need for these techniques to bring quantum teleportation to an acceptable level of data loss for practical applications.

2. Entanglement Swapping

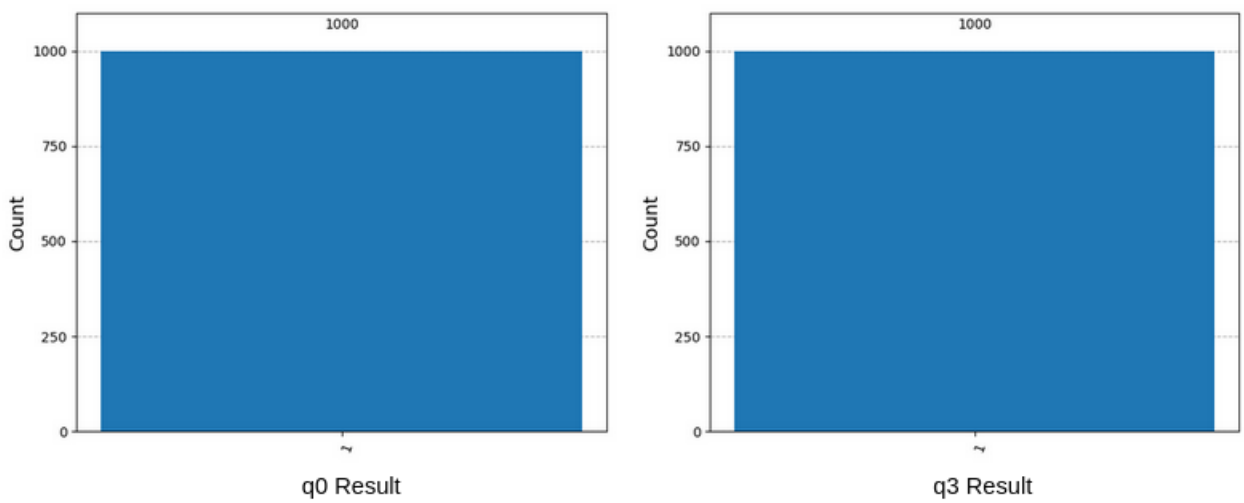
In the second part of the Lab II you will test entanglement swapping. The main goal is to see whether entanglement between qubits survives after teleportation of one of the qubits of a Bell pair.

You will create Bell pair of qubits q_0 and q_1, then teleport qubit q_1 to qubit q_3, and check whether qubits q_0 and q_3 form a Bell pair after the teleportation.

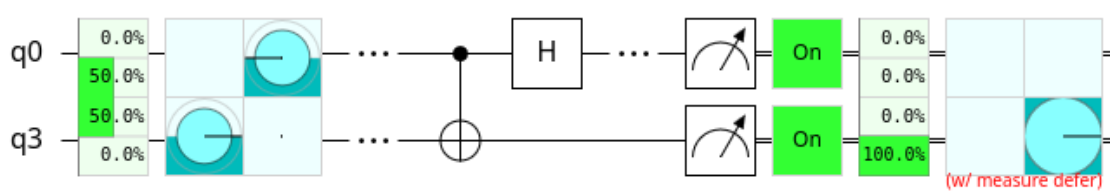
First, we create our quantum circuit to perform the entanglement swapping as shown:



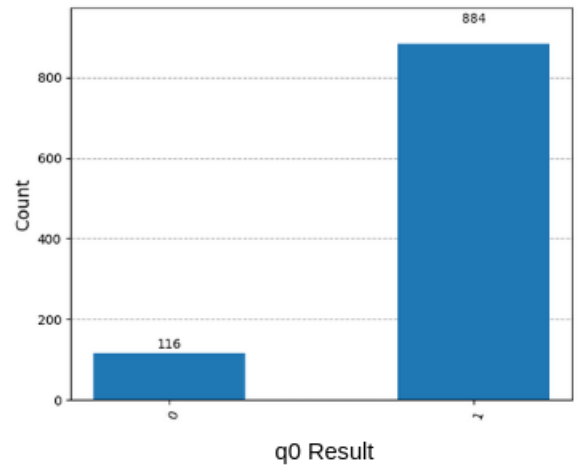
Then we run the circuit on the simulator to theoretically predict the outcome of this process. The final probability results for q_0 and q_3 were:

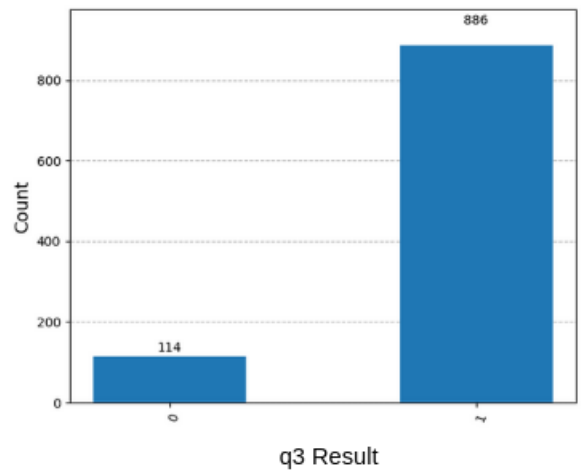


which translates to a theoretically achieved 100% probability of $|q_3 q_0\rangle = |11\rangle$. To illustrate the importance of this outcome to the primary objective, observe in Quirk that this measurement result corresponds to the Bell state $|\Psi^-\rangle$, which would suggest that this circuit should successfully entangle qubits q_0 and q_3:

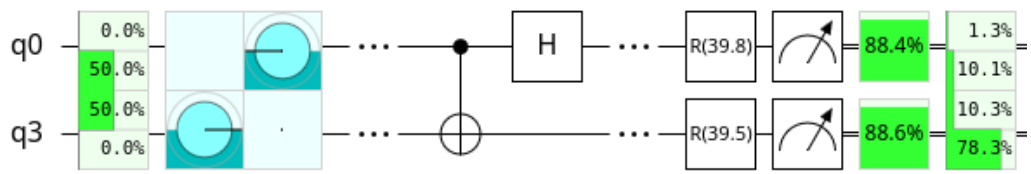


Now that we have our theoretical predictions, we can submit the circuit to an IBM Quantum processor. The results were as follows:





Using custom rotation gates (to represent errors and noise) and the Quirk scheme from above, we can determine the fidelity results of this process on a real quantum processor:



So, we have a fidelity of 78.3% for our uncorrected entanglement swapping process, which falls within the expected range of approximately 60-85% [3] for contemporary experimentation (data range based on photonic quantum processor experiments). Additionally, we observe that the probabilities of either qubit q_0 or q_3 being in the 'ON' state are identical, which suggests that the two qubits are now correlated. We can calculate the level of correlation using density matrices. We can use the density matrix from Quirk for this calculation, so:

$$\rho = \mathbf{Diag}(probabilities)$$
$$\frac{1}{2} \leq \mathbf{Tr}(\rho^2) \leq 1$$

The closer the resulting value is to $\frac{1}{2}$, the more entangled q_0 and q_3 are [4]. Using Pyhton we find that:

```
rho = np.array([[0.013358, 0, 0, 0],
                [0, 0.101942, 0, 0],
                [0, 0, 0.1025, 0],
                [0, 0, 0, 0.7822]])


rho_squared = rho**2
trace = np.trace(rho_squared)
round(float(trace),3)
```

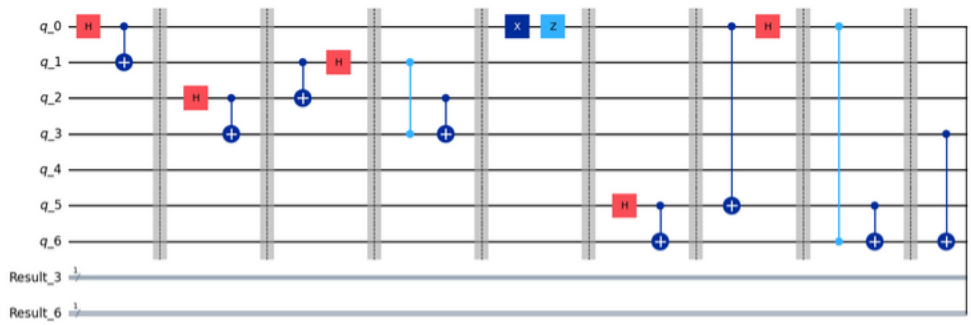
```
0.633
```

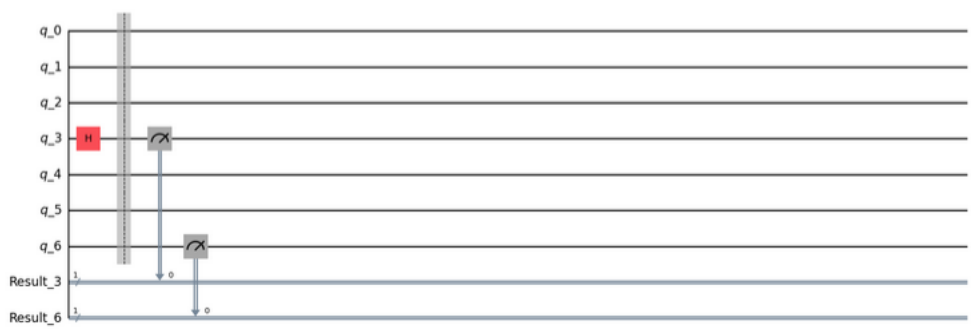
Therefore, we have found that our experimental result has a high level of entanglement, though it is not maximally entangled due to an imperfect fidelity.

3. Bell Pair Teleportation

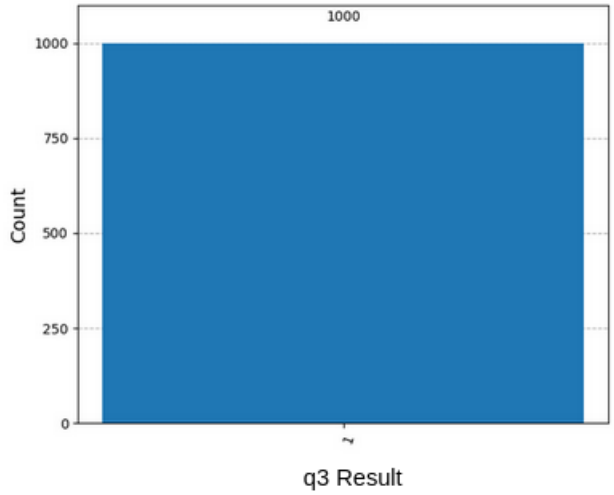
In the second part of the Lab, you teleported only one qubit of a Bell pair. In this part, you will individually teleport both qubits of a Bell pair and check whether they remain entangled and how accurate such "double teleportation" is.

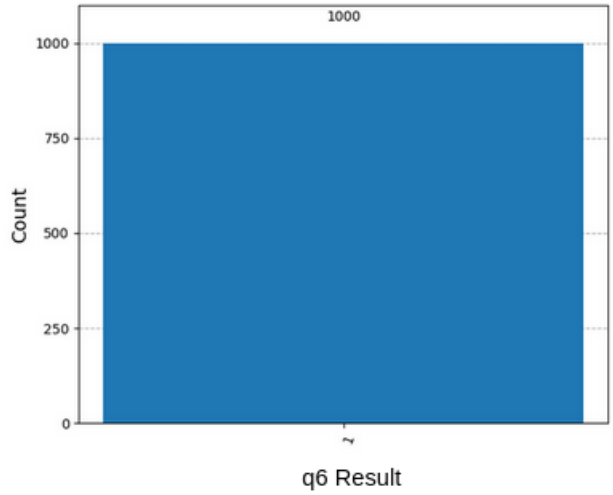
This can be accomplished with a few adjustments to the circuit from the previous section. First, we can use the principle of deferred measurement [5] to simplify the entanglement swapping circuit and apply quantum teleportation to the other qubit (q_0) in the $|\Psi^-\rangle$ state as shown:



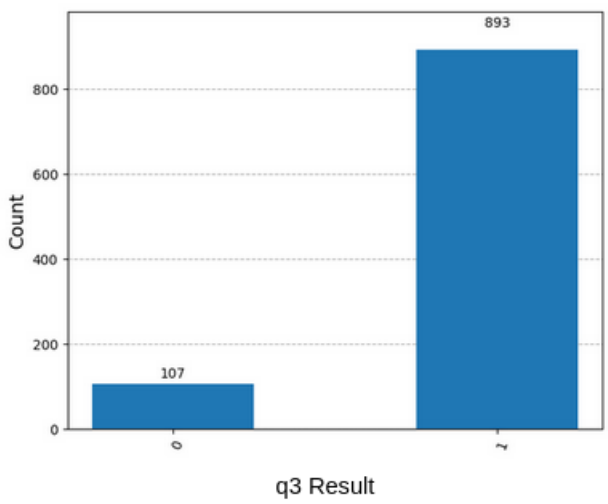


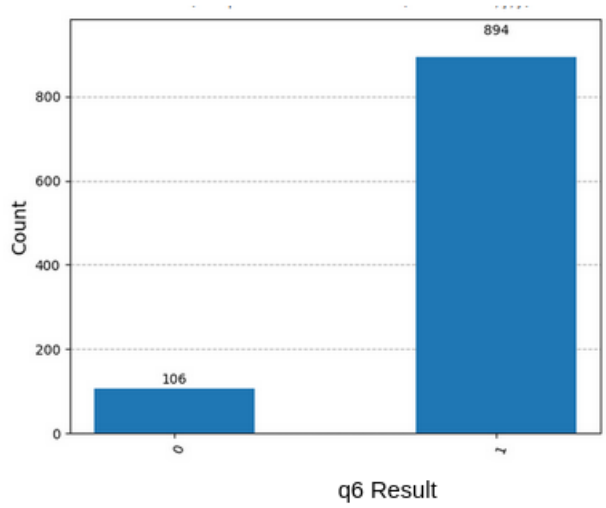
The results of the simulation displayed below are identical to the previous part, so we theoretically should still have entanglement between the teleported qubits.



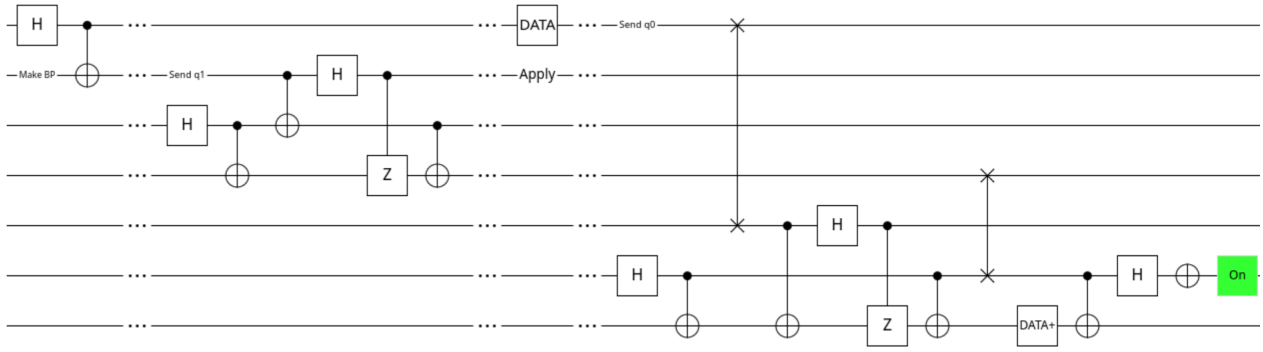


Now we run the circuit on the IBMQ processor, and these were the results:





We observe nearly identical results to the previous part, which suggests that the teleported Bell pair qubits retain their entanglement after being teleported individually with approximately the same fidelity and level of entanglement as in the entanglement swapping case. With this result, we may be able to transfer our data from part 1 by applying it to a qubit in a Bell pair, and after the correlated qubit has been sent to the intended receiver. After data has been applied to the qubit still in the sender's possession, the sender can keep the qubit or send it to someone else for use. Then, the data can be transferred instantly between the two parties in possession of the corresponding paired qubits without having to physically send a qubit encoded with the data, which could be subject to loss. Using quirk to theoretically test this hypothesis, we have:



We observe that qubits of the Bell pair teleported elsewhere have the same state and would likely have a similar fidelity and level of entanglement as results achieved in this section and the previous section. Further experimentation and simulation could garner more insight as to the validaty of this hypothesis.

References

[1] Huang, NN., Huang, WH. & Li, CM. Identification of networking quantum teleportation on 14-qubit IBM universal quantum computer. Sci Rep 10, 3093 (2020). https://doi.org/10.1038/s41598-020-60061-y

[2] Ryan-Anderson, C., Brown, N. C., Baldwin, C. H., Dreiling, J. M., Foltz, C., Gaebler, J. P., ... & Hayes, D. (2024). High-fidelity teleportation of a logical qubit using transversal gates and lattice surgery. Science, 385(6715), 1327-1331.

[3] Tsujimoto, Y., Tanaka, M., Iwasaki, N. et al. High-fidelity entanglement swapping and generation of three-qubit GHZ state using asynchronous telecom photon pair sources. Sci Rep 8, 1446 (2018). https://doi.org/10.1038/s41598-018-19738-8

[4] Nielsen, M. A., & Chuang, I. L. (2011). Quantum Computation and Quantum Information: 10th Anniversary Edition. Cambridge University Press

[5] Wong, T. G. (2022). Introduction to classical and quantum computing. Omaha, NE, USA: Rooted Grove.